

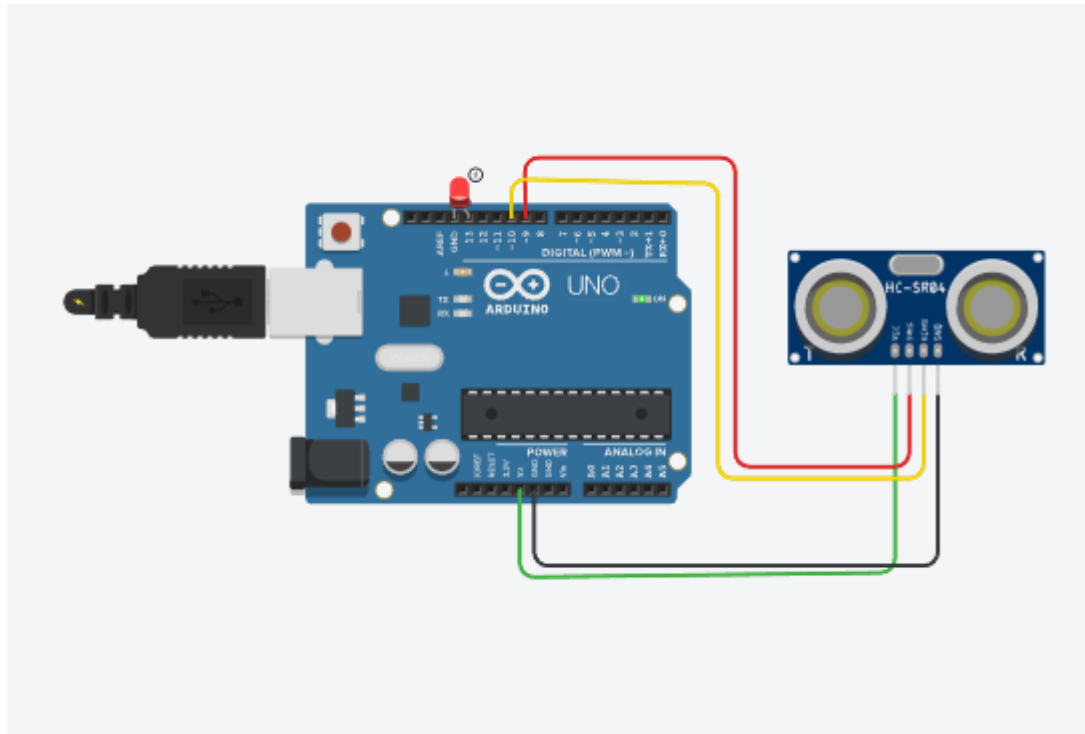
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T11 – 05

Assignment No. 6

Aim: Simulation of sensor-based projects using Tinker Cad

1. ULTRASONIC SENSOR



```

1 // C++ code
2 //
3 const int trigPin = 9;    // Trigger pin for ultrasonic sensor
4 const int echoPin = 10;   // Echo pin for ultrasonic sensor
5 const int ledPin = 13;    // LED pin
6 long totaltime;          // Time taken for the pulse to travel
7 int distance;            // Distance measured
8
9 void setup() {
10   pinMode(trigPin, OUTPUT); // Set the trig pin as an output
11   pinMode(echoPin, INPUT);  // Set the echo pin as an input
12   pinMode(ledPin, OUTPUT);  // Set the LED pin as an output
13   Serial.begin(9600);       // Start serial communication
14 }
15
16 void loop() {
17   digitalWrite(trigPin, LOW); // Ensure the trig pin is low
18   delayMicroseconds(2);       // Wait for a short time
19   digitalWrite(trigPin, HIGH); // Send a pulse
20   delayMicroseconds(10);      // Wait for 10 microseconds
21   digitalWrite(trigPin, LOW); // End the pulse
22
23   totaltime = pulseIn(echoPin, HIGH); // Measure the pulse time on the echo pin
24   distance = totaltime * 0.034 / 2;    // Calculate the distance in cm
25
26   Serial.print("Distance: ");
27   Serial.println(distance);           // Print the distance to the serial monitor
28
29   if (distance > 20) {
30     digitalWrite(ledPin, HIGH); // Turn LED on if distance is more than 20 cm
31     delay(500);                 // Wait for 500 ms
32   } else {
33     digitalWrite(ledPin, LOW);  // Turn LED off if distance is 20 cm or less
34   }
35
36   delay(500);                   // Wait for 500 ms before measuring again
37 }

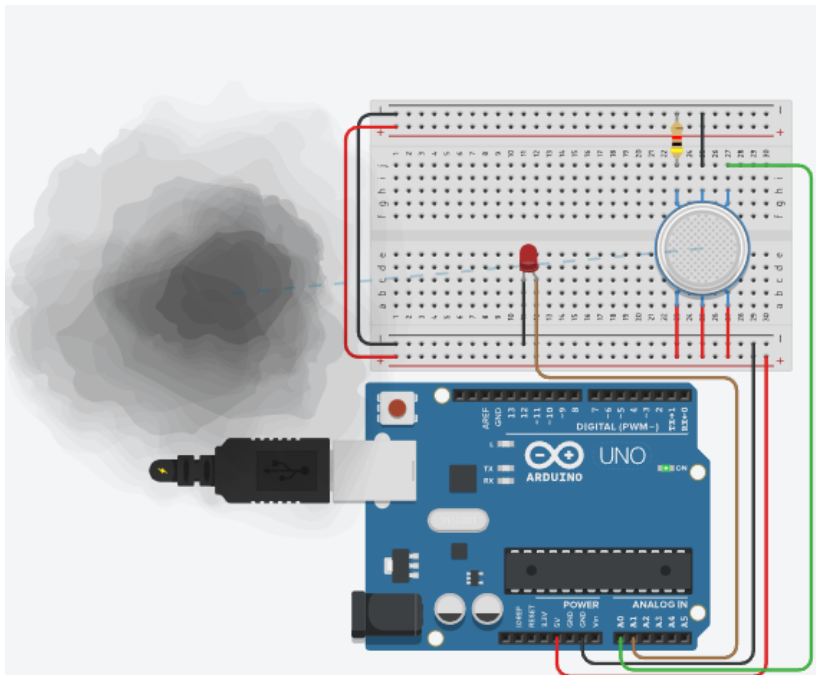
```

```

Distance: 262
Distance: 262
Distance: 262
Distance: 262
Distance: 262
Distance: 262
Distance: 262
Distance: 262
Distance: 118
Distance: 109
Distance: 109
Distance: 22
Distance: 20
Distance: 20
Distance: 20
Distance: 20

```

2. GAS SENSOR



```
1  int LED = A1;
2  const int gas = 0;
3  int MQ2pin = A0;
4
5
6  void setup() {
7      Serial.begin(9600);
8  }
9
10 void loop() {
11     float sensorValue, MQ2pin;
12     sensorValue = analogRead(MQ2pin); // read analog input pin 0
13
14     if(sensorValue >= 470){
15         digitalWrite(LED, LOW);
16         Serial.print(sensorValue);
17         Serial.println(" | SMOKE DETECTED");
18     }
19
20 }
21 else{
22     digitalWrite(LED, HIGH);
23     Serial.println("Sensor Value: ");
24     Serial.println(sensorValue);
25 }
26 delay(1000);
27 }
28
29 float getsensorValue(int pin){
30     return (analogRead(pin));
31 }
```



Serial Monitor

1012.00 | SMOKE DETECTED

Sensor Value:

117.00

Sensor Value:

281.00

743.00 | SMOKE DETECTED

891.00 | SMOKE DETECTED

911.00 | SMOKE DETECTED

767.00 | SMOKE DETECTED

979.00 | SMOKE DETECTED

Sensor Value:

395.00

939.00 | SMOKE DETECTED

870.00 | SMOKE DETECTED

Sensor Value:

111.00

Sensor Value:

340.00

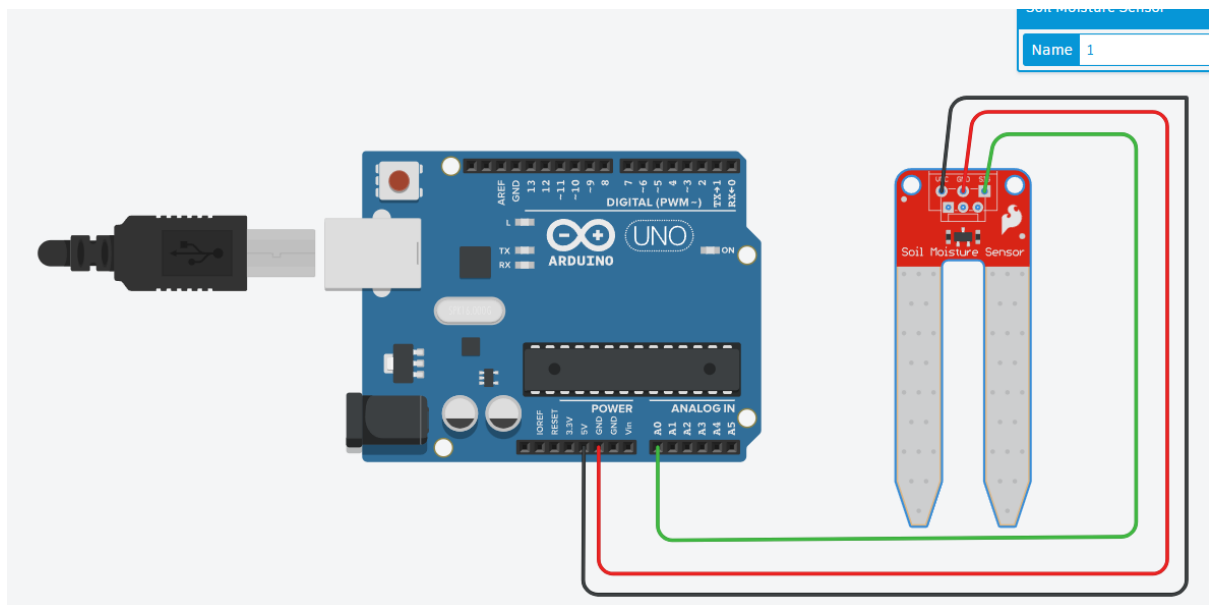
911.00 | SMOKE DETECTED

549.00 | SMOKE DETECTED

945.00 | SMOKE DETECTED

737.00 | SMOKE DETECTED

3. MOISTURE SENSOR



```
1 void setup()
2 {
3     // Set the serial monitor baudrate to 9600
4     Serial.begin(9600);
5 }
6
7 void loop()
8 {
9     // Variable to store ADC value ( 0 to 1023 )
10    int level;
11    // analogRead function returns the integer 10 bit integer (0 to 1023)
12    level = analogRead(0);
13
14    // Print text in serial monitor
15    Serial.println("Analog value:");
16    // Print analog value in serial monitor
17    Serial.println(level);
18 }
19
```



Serial Monitor

0

Analog value:

0

Analog value:

0

Analog value:

0

Analog value:

0

Analog value:

589

Analog value:

589

Analog value:

589

Analog value:

675